

DAY 1 · SATURDAY, 11 JULY 2026



Agentic Software Design & Architecture

BOOTCAMP

Welcome. Let's build engineers who design.

Akın Kaldıroğlu · Emre · Alper Hankendi · Fatih Çakıroğlu

■ The next 60 minutes

This First Hour

1	Welcome & why we're here	10 min
2	Meet your four trainers	10 min
3	The journey & the full schedule	15 min
4	How it works & ground rules	10 min
5	Your turn — introduce yourselves	15 min

...then we dive straight into the first lesson.

Why This Program Exists

Reach developers at the start of their careers — and give them a broad, solid foundation in software design and architecture.



Accessible

A first-of-its-kind bootcamp in the country, priced so students and juniors can actually join.



Foundational

Shallow but wide: the concepts every engineer keeps using — cohesion, coupling, design, architecture.



AI-native

You learn to design and build with AI as a partner — while you stay the decision-maker.

Who's Here · What You'll Gain

Who's in the room

- Students & new graduates
- Junior backend developers (1–2 years)
- Experienced developers who never had formal design training

By the end, you'll be able to

- Reason about cohesion, coupling & change
- Apply Clean Code, SOLID & core design patterns
- Design REST / gRPC APIs and microservices
- Weigh architectural quality attributes & trade-offs
- Lead design & coding with AI — as the decision-maker

SECTION 01

Meet Backendguru and Your Trainers

Four practitioners — one journey from clean code to cloud architecture.

Can Yasun

Backendguru

- Who is Backendguru?
- <https://www.backendguru.com/>

[***https://www.linkedin.com/in/canyasun/***](https://www.linkedin.com/in/canyasun/)

Akın Kaldıroğlu

TEACHES

- Introduction to SW Design & Architecture
- Fundamentals of SW Design · Clean Code · SOLID
- Design Patterns · Secure Coding · Developer Testing
- Agentic Design & Coding · SW Design Project with AI
- Fundamentals of SW Architecture
 - Layered · Hexagonal · Onion · Clean Architecture

<https://github.com/kaldiroglu>

<https://www.linkedin.com/in/akinkaldiroglu/>

My Bootcamp Repos

All the course code lives on GitHub — clone it, run the tests, break it, refactor it. You can send your PRs too.

Companion code · Java *smell → fix, with tests*

▶ github.com/kaldiroglu/Agentic-Software-Design-and-Architecture-Bootcamp-JAVA

Companion code · C# / .NET *smell → fix, with tests*

▶ github.com/kaldiroglu/Agentic-Software-Design-and-Architecture-Bootcamp-DOTNET

Companion code · Python *smell → fix, with tests*

▶ github.com/kaldiroglu/Agentic-Software-Design-and-Architecture-Bootcamp-PYTHON

Refactoring exercises *3 projects (C#, Java, Python) for the hands-on exercises*

▶ github.com/kaldiroglu/refactoring-course-student-materials

Emre Akış

TEACHES

- Introduction to Distributed Architecture: SOA & Microservices
- API Design & Microservices
- From monoliths to distributed systems — risks & benefits

<https://github.com/emreakis>

<https://www.linkedin.com/in/emreakis/>

Alper Hankendi

TEACHES

- **Data Architecture: Event Sourcing**
- **System Design Project with AI · Spec-driven development**
- **System Design Interview — how to prepare & perform**

Fatih Çakıroğlu

TEACHES

- **Security Architectures** — secure app development & web security
- **Cloud Architectures** — AWS · Google Cloud · Microsoft Azure

The Journey — Three Phases

PHASE 1

Software Design

Small-scale · foundations · ~24 h

- Design fundamentals · cohesion & coupling
- Clean Code · SOLID · Design Patterns
- Secure Coding · Object Stereotypes · Developer Testing
- Agentic coding + a design project with AI

PHASE 2

Software Architecture

Large-scale · systems · ~22 h

- Architecture fundamentals & models (4+1, C4)
- Layered · Hexagonal · SOA · Microservices
- API design · Data & Event Sourcing
- Security · Cloud · System design project & interview

PHASE 3

Project Presentations

Every team presents and defends its design · live Q&A · ~4 h

Schedule — Phase 1: Software Design

DATE	TOPIC	TRAINER
Sat 11 Jul	Introduction to SW Design & Architecture, Fundamentals of SW Design	Akın
Tue 14 Jul	Fundamentals of SW Design	Akın
Sat 18 Jul	Fundamentals of SW Design & Clean Code	Akın
Tue 21 Jul	Clean Code	Akın
Sat 25 Jul	SOLID & Design Patterns	Akın
Wed 29 Jul	Secure Coding	Akın
Sat 1 Aug	Object Stereotypes & Developer Testing	Akın
Wed 5 Aug	Agentic Design & Coding + SW Design Project with AI	Akın

Today: after this welcome hour, we begin 'Introduction to SW Design & Architecture'.

Schedule — Phase 2: Software Architecture

DATE	TOPIC	TRAINER
Sat 8 Aug	Fundamentals of SW Architecture, Layered & Hexagonal/Onion/Clean Architecture	Akın
Wed 12 Aug	Intro to Distributed Architectures: SOA & Microservices	Emre
Sat 15 Aug	Microservices & API Design & Microservices	Emre
Wed 19 Aug	No class today!	
Sat 22 Aug	Security & Cloud Architectures	Fatih
Wed 26 Aug	Data Architecture: Event Sourcing	Alper
Sat 29 Aug	System Design Project with AI	Alper
Wed 2 Sep	System Design Interview	Alper

Schedule — Phase 3: Project Presentations

DATE	TOPIC	TRAINER
Sat 12 Sep	Project Presentations	Everyone

How It Works

Cadence

Saturdays 10:00–14:00 (4 h) · Wednesdays 20:00–22:00 (2 h)

Duration

~50 hours over 8 weeks (Jul–Aug) · presentations 12 Sep

Languages

Examples in C#, Java & Python

AI-first

We work with Claude Code — prompting & conversation techniques; you stay the decision-maker

Capstone

A real business problem, in small groups (4–5) · 10–15 min final presentations

Everything on GitHub

Code, exercises and projects are shared and reviewed on GitHub

Your Certificates

You leave the bootcamp with proof of your work — and finishing your project earns you more.

EVERY PARTICIPANT

Certificate of Participation

Every participant in the bootcamp receives a Certificate of Participation.

PROJECT + PRESENTATION

Certificate of Completion

Complete your capstone project and present it to earn a second certificate recognizing your completed project — in addition to your participation certificate.

Ground Rules & Logistics

01 Show up

Be present and on time — Saturdays 10:00, Tuesdays 20:00.

02 Speak up

Cameras and mics welcome — this is a conversation, not a broadcast.

03 Ask anything

There is no such thing as a dumb question here.

04 Practice

Work between sessions — design is a muscle you build by using it.

05 Recordings

Sessions may be published to our YouTube channel afterward.

06 Stay connected

Community channel: [fill in — Discord / WhatsApp / Telegram]

07 Stay informed

Please take note of all important dates and links, and pay close attention to messages received via email or the WhatsApp channel.

SECTION 02

Your Turn — Introduce Yourself 🙌

Tell us who you are — keep it short so everyone gets a turn.

■ Your turn

Look! Who is here?

Why are you here?

- Who wants to introduce themselves, and tell us why they are here?
- What are your expectations from this bootcamp?
- What makes this bootcamp perfect for you?

Let's get connected: please share your LinkedIn profile in the WhatsApp channel.

Right after a short break

What's Next?

LESSON 1 · TODAY

Introduction to Software Design & Architecture

with Akin Kaldıroğlu

What design is, what architecture is — and why they are not the same thing.

Grab water, stretch — back in 15 minutes.

Let's Begin.

Eight weeks. From clean code to cloud. Let's design.

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